

New York University

Tandon School of Engineering

Department of Computer Science and Engineering

Introduction to Operating Systems Fall 2026

Assignment 3 (10 points)

Develop a simple Linux kernel module that runs on your virtual machine. The only functionality required of your module is to be able to load and unload, printing a debug message while doing so.

When a Linux kernel module is loaded, it invokes an init function, and when it is removed (or unloaded), it invokes an exit function.

- A) (0 points) Read chapter 2 of the freely available O'Reilly book "Linux Device Drivers, 3rd Edition" (<https://lwn.net/Kernel/LDD3/>), in particular p.16, as well as your text book p.96 to get you started. Note that even though the LDD3 book is written for kernel version 2.6, most mechanisms are applicable with minor or no changes. The relevant example code is copied below as a starting point.

```
#include <linux/init.h>
#include <linux/module.h>
MODULE_LICENSE("Dual BSD/GPL");
static int hello_init(void)
{
    printk(KERN_ALERT "Hello, world\n");
    return 0;
}
static void hello_exit(void)
{
    printk(KERN_ALERT "Goodbye, cruel world\n");
}
module_init(hello_init);
module_exit(hello_exit);
```

The `hello_init()` function is invoked when you insert your module (using the `insmod` shell command), whereas the `hello_exit()` is called when you unload your module (using the `rmmod` shell command).

- B) (0 points) Read the description of the global kernel variable `jiffies` and the macro `HZ` in the O'Reilly book (search in the searchable pdf). Then read about the `ktime_get_boottime()` routine in <https://www.kernel.org/doc/html/latest/core-api/timekeeping.html>

- C) (0 points) You may need to install the kernel headers if not already installed. Type:

```
sudo apt-get install linux-headers-$(uname -r)
```

- D) (10 points) Modify the c code given above (which becomes your `lab3.c`) such that:

- 1) The init function prints a "hello" c-string, followed by the tick time in milliseconds (i.e. the timer interval, as we defined it in weeks 1/2).

The init function shall also obtain the current time (see the hint below) and then save the value of `jiffies` and the current time using `ktime_get_boottime(void)` (both saved in static global variables, obviously).

Hint: Further info may be found in <https://docs.kernel.org/core-api/timekeeping.html>

- 2) The exit function reads the timer again (yes, using `ktime_get_boottime(void)`, again) and prints a “goodbye” c-string, followed by the time difference **in milliseconds** between the insertion and removal of the module i.e. between init and exit functions) using two different methods:
 - a. Using the difference in the value of `jiffies` from inserting the module to removing the module. Of course the `jiffies` are not time units (milliseconds), so you need to convert them to time, based on your proper understanding of the topic.
 - b. Using the time difference obtained by reading the timer (Hint: use `ktime_get_boottime()`).

You shall use the Makefile provided with the assignment (In some cases, you may need to slightly modify the Makefile provided to suit your setup). You should place it in the same directory as your `.c` file (lab3.c)

Hints:

- E) Your module should use `printk()` to print messages. You will use this print facility to also debug your code if needed (😊). More information may be found on <https://www.kernel.org/doc/html/latest/core-api/printk-basics.html>
- F) Use `dmesg` shell command to view messages printed by `printk()`, e.g. type:

```
dmesg
```

You may clear the log using:

```
dmesg -C
```

What to submit to [gradescope](#):

Please submit the following files individually:

- 1) Source file(s) with appropriate comments.
The naming should be similar to “**lab#_\$.c**” (`#` is replaced with the assignment number and `$` with the question number within the assignment, e.g. `lab4_b.c`, for lab 4, question b OR `lab5_1a` for lab 5, question 1a).
- 2) A single pdf file (for images + report/answers to questions), named “**lab#.pdf**” (`#` is replaced by the assignment number), containing:
 - Screen shot(s) of your terminal window showing the current directory, the command used to compile your program, the command used to run your program and the output of your program.
- 3) Your Makefile, if any. This is applicable only to kernel modules.

RULES:

- You shall **use kernel version 4.x.x or above**. You shall not use kernel version 3.x.x.
- You may consult with other students about GENERAL concepts or methods but copying code (or code fragments) or algorithms is NOT ALLOWED and is considered cheating (whether copied from other students, the internet or any other source).
- If you are having trouble, please ask your teaching assistant for help.
- You must submit your assignment prior to the deadline.